4.4 Surface Area; Parametric Surfaces Worksheet

1. Consider the surface σ given by $x+y+2z=2$ which lies in the first octant. Determine the exact surface area of σ.

2. Consider the surface σ given by $z = 3x^2 + 3y^2$ which lies below the plane $z = 27$. Determine the exact surface area of σ.

3. Consider the surface determined by $f(x, y) = \sin(y)$ where $0 \le y \le 2\pi$ and $-1 \le x \le 1$.
   a. Set up an iterated integral which computes the surface area of this surface.

   b. Find a parameterization of the surface.

c. Set up an iterated integral which computes the surface area of this parameterized surface.

4. Consider the part of the graph $z = 1 - x^2$ where $z \geq 0$ and $-2 \leq y \leq 2$.

a. Set up an iterated integral which computes the surface area of this surface.

b. Find a parameterization of the surface.

c. Set up an iterated integral which computes the surface area of this parameterized surface.

5. Consider the surface σ with the parameterization $\mathbf{r}(u,v)=\langle u,1-u,v\rangle$ with $0\le u\le 1$ and $0\le v\le 2$.

a. Sketch a picture of σ and determine its surface area algebraically.

b. Set up an iterated integral which computes the surface area of σ and confirm that it evaluates to the surface area you found in part (a).

c. Can this surface be presented as a function $z = f(x, y)$?

6. Consider the surface σ with the parameterization $\mathbf{r}(u,v) = \langle v\cos(u), v, v\sin(u)\rangle$ with $0 \le u \le 2\pi$ and $0 \le v \le 1$.

   a. Sketch a picture of σ and describe its shape.

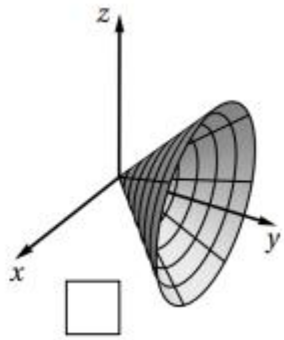
   b. Set up an iterated integral which computes the surface area of σ and calculate this integral.

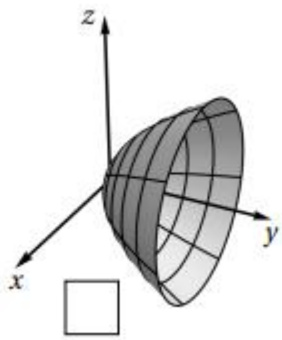
c. Find a way to write express σ as *two* functions of the form $z = f(x, y)$

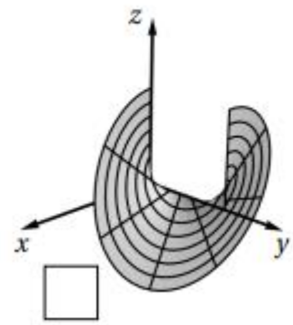
d. Determine the surface area of σ using functions you found in part (c) and confirm that your result matches the result from part (b).

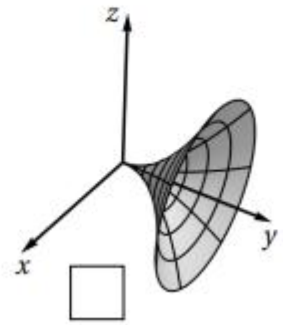
7. Consider the surface σ parameterized by $\mathbf{r}(u,v)=\langle u^2\sin(v),u,u^2\cos(v)\rangle$ for $0\le u\le 1$ & $0\le v\le 2\pi$,

a) Mark the box below which corresponds to the picture of σ.





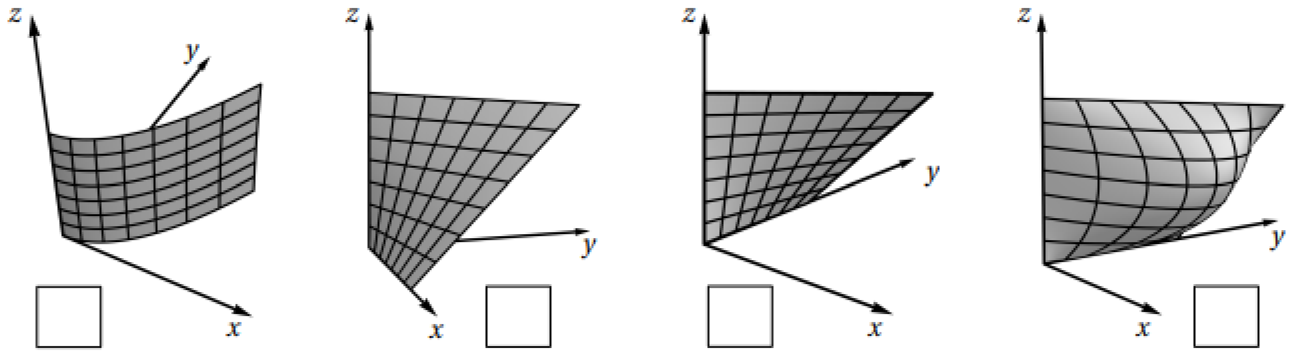




b) Set up an iterated integral which computes the surface area of σ.

8. Consider the surface σ parameterized by $\mathbf{r}(u,v) = \langle uv, u, v \rangle$ for $0 \le u \le 1$ & $0 \le v \le 1$,

a) Mark the box below which corresponds to the picture of σ.



b) Set up an iterated integral which computes the surface area of σ.

c) Express $\mathbf{r}(u,v)$ as a function in terms of x, y, & z.

9. Consider a cylindrical can of radius 4 and a height of 5 (including top and bottom).

a) Create a parameterization of these surfaces, you will need 3 parameterizations in total. Note: Infinitely many parameterizations exist.

b) Set up iterated integrals to compute the surface area of each of the three surfaces from part (a) to determine the surface area of the entire can.